



LABORATORY ACTIVITY NO. 3.1.11

Specifying PC Systems

This lab activity is designed to familiarize the student with the costs and specifications associated with various types of personal computers, including desktops, laptops, netbooks, and handheld devices. They will be tasked with using online vendor stores, such as HP (hp.com) and Dell (dell.com), to research and obtain system specifications for different PC requirements based on predefined budgets. The lab also encourages you to compare various technologies, like CPU and graphics card models, once you have gathered the specifications.

I. OBJECTIVES

Upon completion of this laboratory exercise, students are expected to and meet the following intended learning outcomes:

Activity Objectives	Intended Learning Outcomes		
	ILO1	ILO2	ILO3
1) effectively utilize online vendor stores (e.g., HP, Dell) to research, identify, and configure PC systems (desktop, laptop, netbook) that meet specific performance requirements and adhere to predefined budget constraints.	•	•	
2) compare and contrast competing technologies and products from various component manufacturers (e.g., CPUs, graphics cards) to understand their specifications, performance characteristics, and impact on overall system design.	•	•	
3) develop an understanding of the market landscape for PC systems, enabling the selection of appropriate components and configurations to satisfy diverse user needs and budget considerations for different types of computers.	•	•	

Where:

- ILO1 – Exhibit thorough demonstration of the concept
- ILO2 – Being able to perform the activity independently
- ILO3 – Design and evaluate emerging IT solutions



II. DISCUSSION

- **Selecting a Computer**

- Key components affecting performance include the Central Processing Unit (CPU), Memory (System RAM), Fixed disk (HDD or SSD), and Graphics Processing Unit (GPU).

- **Network Interface**

- Covers wired network connections (Ethernet port, Network Interface Card - NIC) and wireless networks (Wi-Fi radio, Wireless access points).

- **Motherboard Components**

- The motherboard is a Printed Circuit Board (PCB) containing processors (chipset), sockets for upgradable components (CPU, RAM, adapter cards, disk drives), and buses that connect them.
- It dictates the computer's upgrade potential.
- Key components on a motherboard include the CPU Socket, Memory Slots, Disk Drive Connectors, Adapter Card Slots, and I/O Ports (USB, video, audio, etc.).

- **Processors**

- The CPU is the main processor, with major manufacturers including Intel and AMD.
- Intel CPU brands include Core, Pentium, Celeron, Atom, and Xeon, while AMD features Ryzen/Threadripper and Epyc, based on Zen microarchitecture.
- ARM microarchitecture is widely used in smartphones and tablets, with companies like Apple, Samsung, and NVIDIA licensing ARM designs.
- Discusses RISC (Reduced Instruction Set Computing) versus CISC (Complex Instruction Set Computing) architectures, noting Intel's CISC design with RISC enhancements.
- Processor features include the control unit, execution units, and registers.

- **Instruction Set (32-bit versus 64-bit)**

- The x86 instruction set evolved from 16-bit to 32-bit (x86-32/IA-32) and then to 64-bit (x86-64/x64).
- 32-bit CPUs can address up to 4 GB of memory, while 64-bit CPUs can address significantly more (currently up to 256 Terabytes using a 48-bit address space).



- 64-bit CPUs can run both 32-bit operating systems and software, but 32-bit CPUs cannot run 64-bit operating systems or software. Most PC/laptop CPUs are now 64-bit.
- **Clock Speed and Bus Speed**
 - Clock speed is a key differentiator for CPU models, with premium models running faster.
 - The Front Side Bus between the CPU and system memory is another important performance factor.
- **Multiprocessing**
 - **Symmetric Multiprocessing (SMP)** involves multiple CPU sockets on the motherboard, typically a feature of server systems.
 - **Chip Level Multiprocessing (CMP)** means a CPU contains multiple "cores," each acting as an independent CPU (e.g., 2-core, 4-core, 8-core models).
- **System and Expansion Bus Technologies**
 - The motherboard uses buses to connect components, transferring data, address information, timing signals, and power.
 - The **System bus (Front Side Bus)** connects the CPU and system memory.
 - The **Expansion bus (Input/Output Bus)** connects the CPU to other components.
 - Provides an overview of various Expansion Bus Types.
- **System Cooling**
 - **Heatsinks and thermal paste** provide passive cooling by transferring heat through convection.
 - **Fans** offer active cooling, dissipating warm air from components and moving air through the case.
 - **Liquid-based Cooling Systems**
- **BIOS and UEFI System Firmware**
 - **Bootstrapping (booting)** is the process where the CPU initializes components before the operating system loads, controlled by PC firmware.
 - **BIOS (Basic Input/Output System)** and **UEFI (Unified Extensible Firmware Interface)** are types of PC firmware.
 - Describes accessing the system firmware setup program by pressing a key during the boot sequence and navigating using the keyboard.



III. MATERIALS

A Windows 11 installed computer system.

IV. PROCEDURE

For each of the PC system requirements listed below, you are to conduct thorough online research to identify suitable configurations that meet the specified budget. Your research should be comprehensive and comparative, ensuring you explore various options before making your selection.

1. **Vendor Research:** Choose at least **two different computer vendors** (e.g., HP, Dell, Lenovo, Acer, etc.) and explore their online stores to configure or identify pre-built systems that align with the given requirements and budget. You can start with vendors such as HP (hp.com) and Dell (dell.com) as suggested in the lab.
2. **Component Manufacturer Identification:** For each system you identify, determine the primary manufacturers of at least two key components (e.g., CPU, Graphics Card). For example, if you find a system with an Intel Core i7 processor, Intel is the manufacturer.
3. **Technology Comparison:** Research at least **two competing technologies** for a key component (e.g., compare an Intel CPU with an AMD CPU, or an NVIDIA graphics card with an AMD Radeon graphics card). Note down the models and key specifications of these competing technologies to understand their differences.
4. **Specification Gathering:** Obtain detailed specifications for the components of your chosen system, including processor model, RAM capacity and type, storage type and size, graphics card model, and display specifications where applicable.
5. **Budget Adherence:** Ensure that your selected system's estimated cost falls within the specified budget, excluding sales taxes and delivery charges.
6. **Documentation:** Fill in the blank table provided below with the specifications you gather. This table will help you organize your findings and facilitate comparisons.

Important Considerations:

- **Vendor Research:** Utilize the online stores of reputable vendors like HP and Dell to build or find pre-configured systems that meet the specified requirements.
- **Budget Adherence:** Strictly adhere to the provided budget for each PC type. Remember that all budgets are to exclude sales taxes and delivery charges. The requirements include:
 - A tower case business system (without a monitor) for P20,000.00.
 - A portable laptop or netbook for basic word processing, email, and web access for P25,000.00.



- An ultraportable business laptop with a fast CPU and sufficient memory for P40,000.00.
- A desktop media center/gaming PC featuring high-specification graphics and a premium HD widescreen flat-panel display for P60,000.00.
- **Detailed Specifications:** Focus on identifying key components and their specifications, such as CPU type and speed, amount of RAM, storage type and size (SSD/HDD), graphics card model, and display features, to accurately match the requirements.
- **Technology Comparison:** After gathering specifications for the required systems, take the time to compare the underlying technologies, such as different CPU models or graphics cards, to understand their relative performance and value.

Preparation:

**Please fill out this table for each of the required PC systems. You may also wish to alter the format of the table for better presentation.



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PC System Specifications Research Table

PC System Requirement	Vendor 1 (Name & URL)	Vendor 1 - Key Specs (CPU, RAM, Storage, GPU, Display)	Vendor 1 - Estimated Cost (in Pesos)	Vendor 2 (Name & URL)	Vendor 2 - Key Specs (CPU, RAM, Storage, GPU, Display)	Vendor 2 - Estimated Cost (in Pesos)	Component Manufacturer 1 (e.g., CPU)	Competing Technology 1 (Manufacturer & Model)	Component Manufacturer 2 (e.g., GPU)	Competing Technology 2 (Manufacturer & Model)
Tower case business system (no monitor, P20,000.00)	Dell Vostro 3020 Tower, dell.com/en-ph	Intel Core i3-13100 (4C/8T, up to 4.5 GHz), 8 GB DDR4, 256 GB SSD, Intel UHD Graphics 730	P19,990.00	HP Pro Tower 290 G9, hp.com/en-ph	Intel Core i3-12100 (4C/8T, up to 4.3 GHz), 8 GB DDR4, 256 GB SSD, Intel UHD Graphics 730	P19,500.00	Intel	AMD Ryzen 5 5600G (6C/12T, up to 4.4 GHz)	Intel (Integrated)	AMD Radeon Graphics (7-core Vega, integrated)
Portable (laptop or netbook) for basic word processing, email, web access (P25,000.00)	HP 15-fd Laptop, hp.com/en-ph	Intel Core i3-N305 (8 cores, up to 3.8 GHz), 8 GB DDR4, 256 GB SSD, Intel UHD Graphics, 15.6" FHD (1920x1080)	P24,990.00	Dell Inspiron 15 3530, dell.com/en-ph	Intel Core i3-1305U (5C/6T, up to 4.5 GHz), 8 GB DDR4, 512 GB SSD, Intel UHD Graphics, 15.6" FHD (1920x1080), 120 Hz	P24,500.00	Intel	AMD Ryzen 3 7320U (4C/8T, up to 4.1 GHz)	Intel (Integrated)	AMD Radeon 610M (integrated)



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Ultraportable business laptop with fast CPU and memory (P40,000.00)	HP ProBook 440 G10, hp.com/ph-en	Intel Core i5-1335U (10C/12T, up to 4.6 GHz), 16 GB DDR4, 512 GB NVMe SSD, Intel Iris Xe Graphics, 14" FHD IPS, about 1.4 kg	P39,900.00	Dell Latitude 3450, dell.com/en-ph	Intel Core i5-1335U (10C/12T, up to 4.6 GHz), 16 GB DDR4, 512 GB NVMe SSD, Intel Iris Xe Graphics, 14" FHD anti-glare	P39,500.00	Intel	AMD Ryzen 5 7530U (6C/12T, up to 4.5 GHz)	Intel (Integrated)	AMD Radeon Graphics (Vega 7, integrated)
Desktop media center/gaming PC with high-spec graphics and premium HD widescreen flat-panel display (P60,000.00)	HP Victus 15L Gaming Desktop + HP 27" monitor, hp.com/ph-en	Intel Core i5-13400F (10C/16T, up to 4.6 GHz), 16 GB DDR4, 512 GB NVMe SSD, NVIDIA GeForce RTX 4060 8 GB GDDR6, 27" FHD IPS widescreen, 165 Hz	P57,500.00 (PC P48,000 + monitor P9,500)	Acer Nitro 50 + Acer 24" 165 Hz monitor, acer.com/ph-en	Intel Core i5-14400F (10C/16T, up to 4.7 GHz), 16 GB DDR5, 512 GB NVMe SSD, NVIDIA GeForce RTX 4060 8 GB GDDR6, 24" FHD IPS widescreen, 165 Hz	P57,000.00 (PC P50,000 + monitor P7,000)	Intel	AMD Ryzen 5 7600 (6C/12T, up to 5.1 GHz)	NVIDIA	AMD Radeon RX 7600 8 GB GDDR6



V. REFLECTION:

Upon successful completion of "Lab 3.1.11: Specifying PC Systems," write a comprehensive reflection (between 150-250 words) addressing the following points:

- **Learning Outcomes:** What new knowledge or skills did you acquire or reinforce regarding PC system specification?
- **Challenges and Solutions:** Describe any difficulties or unexpected issues you encountered during the lab. How did you overcome them, or what steps would you take to resolve them in the future?
- **Real-world Relevance:** How do the skills and understanding gained in this lab relate to real-world scenarios, either in your personal life or a professional IT context?
- **Critical Thinking:** If you had to use an internet image for any step, explain why you couldn't perform the action physically and how finding the image contributed to your understanding.

This lab helped me understand how the CPU, RAM, storage, and graphics work together to match a computer to a user's needs and budget. I learned to read vendor specifications such as core count, clock speed, DDR memory type, SSD capacity, and display resolution, and to compare Intel and AMD processors as well as NVIDIA and AMD graphics cards.

I did not encounter any challenges during this laboratory activity. The vendor websites were easy to navigate, and the specifications were clear enough to fit each system within its budget.

These skills apply directly when buying a computer for school, work, or gaming, and in IT roles where technicians must recommend or procure hardware that meets user requirements without overspending. Knowing which specifications matter for each use case avoids paying for unnecessary features.

I did not use any internet image in this activity. All the information came from the specifications and product details listed on the vendor websites, so no image was needed to perform or explain any step. Comparing the specifications directly helped me understand the performance differences between competing components.